

BIG SYNTAX DOESN'T WANT YOU TO KNOW

MAL is universal? Hardly.

The Menzerath–Altmann law on all 180 UD treebanks

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HOW WE COMPUTE MAL

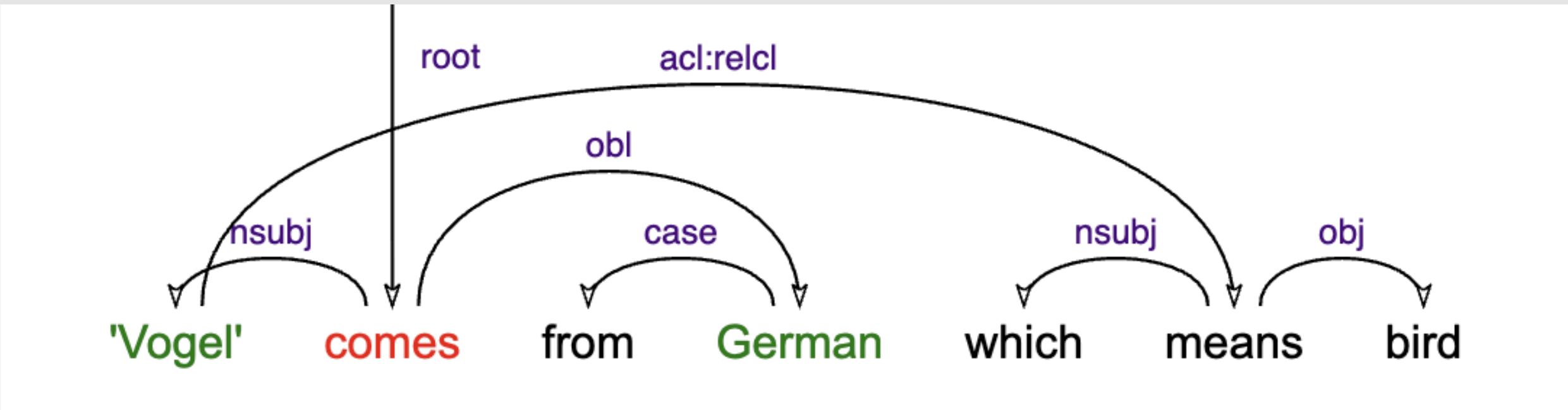
Real constituents only. Fake dependents eliminated!

The **Menzerath–Altmann law** (MAL) predicts that “the greater the number of components of a given unit, the shorter their respective length” (Menzerath, 1928). In the verbal domain: as the number of dependents n grows, their average size decreases.

Altmann (1980) formalised this as a **power law**: $MAL_n \approx n^{-\beta}$, i.e. $\log(MAL_n) \approx -\beta \cdot \log(n)$. The slope β measures the **MAL effect**. Previous studies: Macutek et al. (2017) for Czech; Tanaka-Ishii (2021) for UD 2.3 without filtering.

NEW! Our data extraction is filtered: we exclude punct, discourse, parataxis, conj, cc, vocative, aux, compound:prt, mark, case from the verbal construction. Only **real syntactic constituents** are retained.

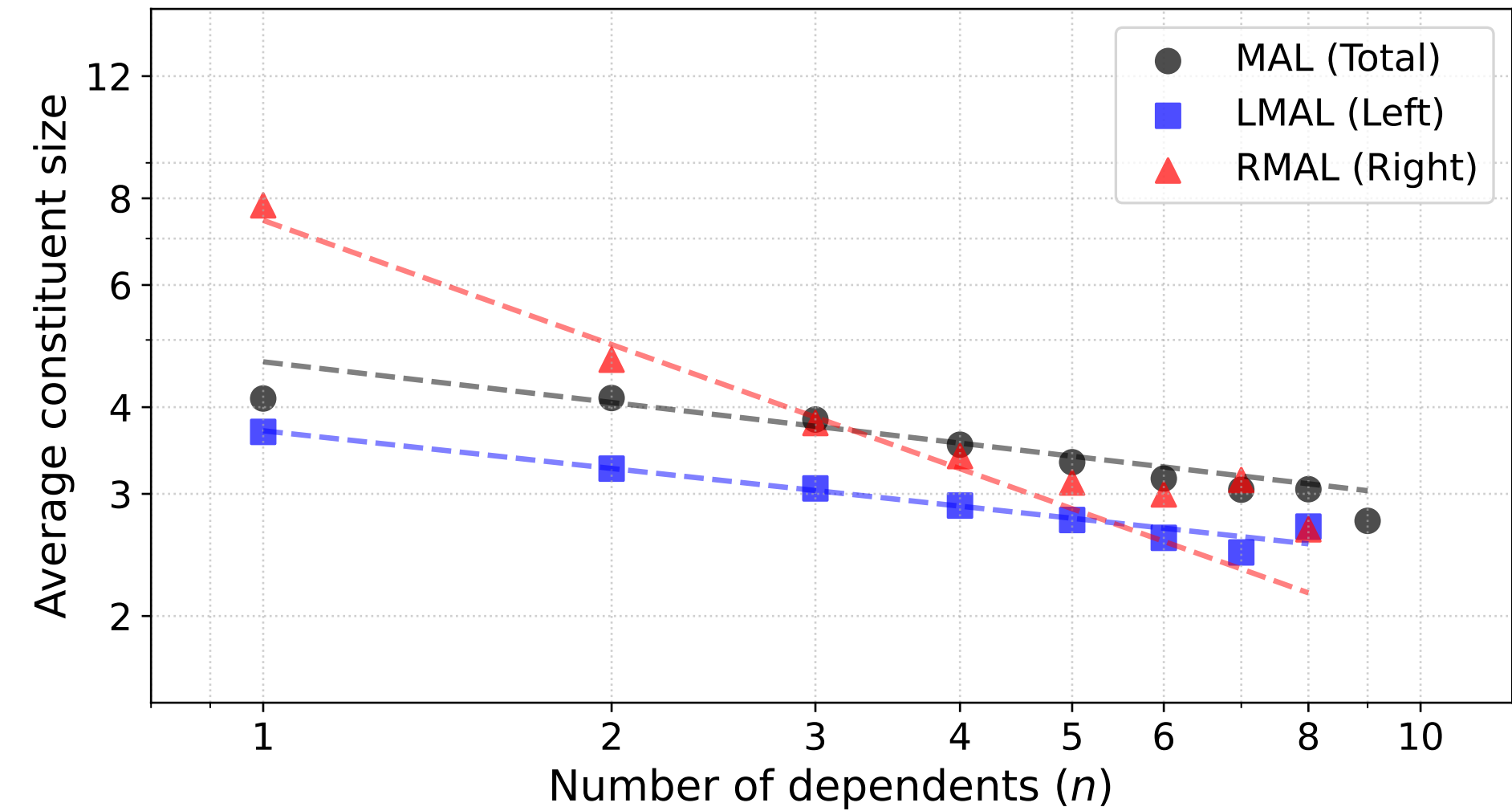
What about “bastards”? A constituent’s subtree may spill across the verb. We count the spilled part as a separate dependent of the verb:



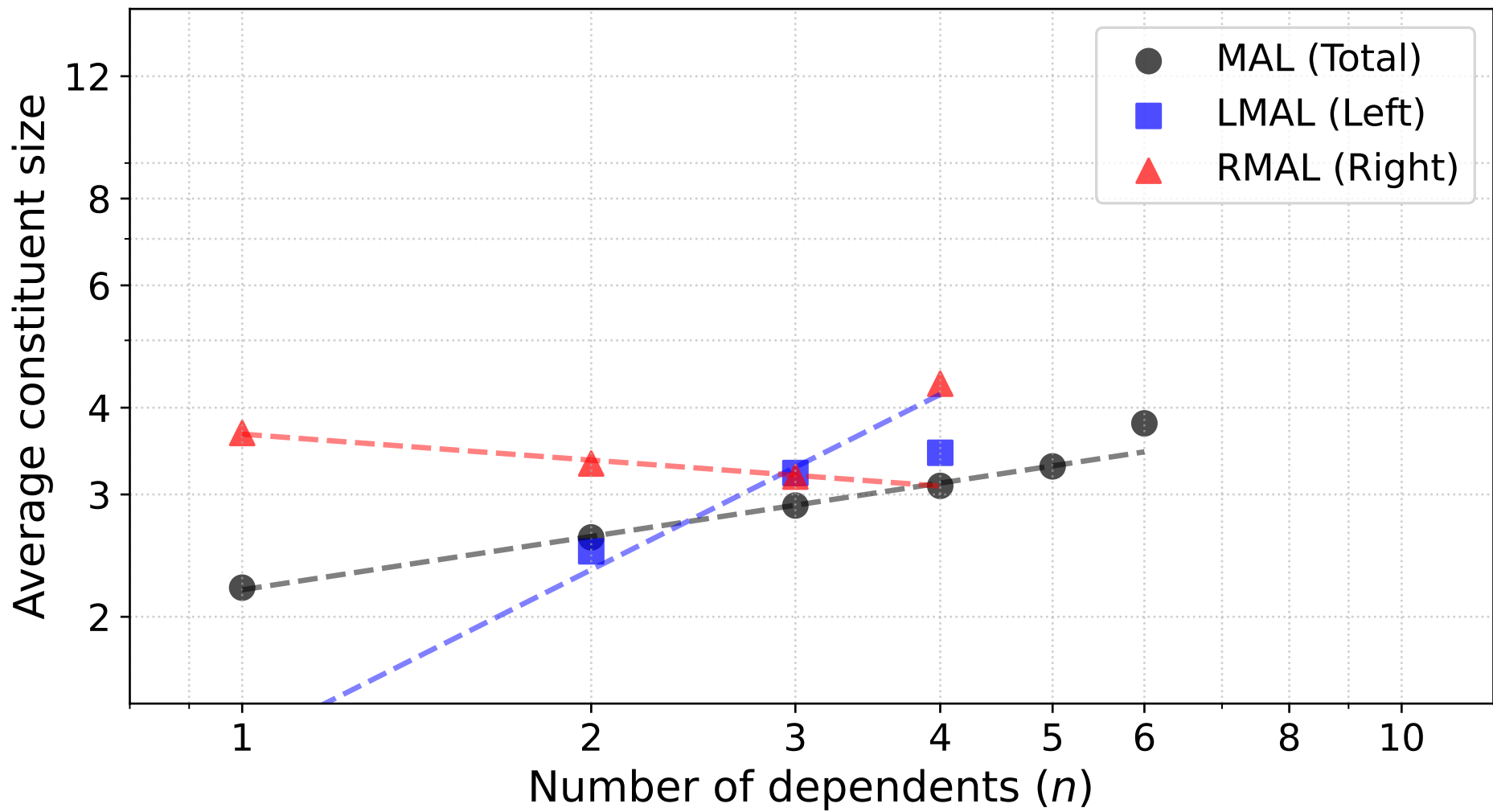
The relative clause “which means bird” (**acl:relcl**) syntactically depends on ‘Vogel’ (left of verb) but appears right of verb – a **bastard**. It is counted as a right-side constituent of **comes**. Source: en_gum.

AS USUAL

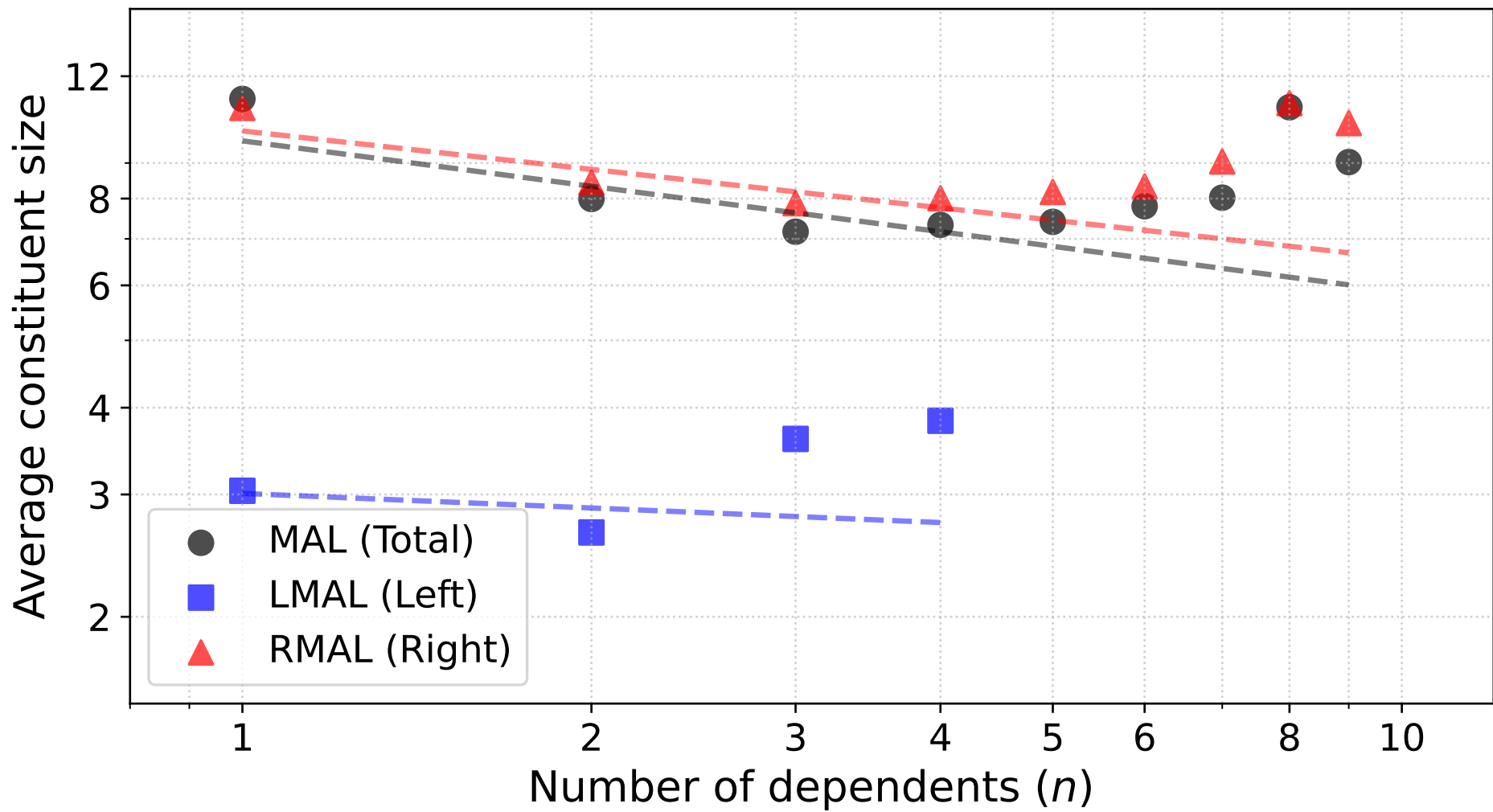
German nails it. Naija wobbles. Arabic surprises. You?



German – textbook MAL
slope = -0.17, R² = 0.89 on the log-log fit. A near-perfect power law: every extra dependent shrinks the average constituent. German is MAL, RMAL and LMAL as expected. Is anyone surprised?



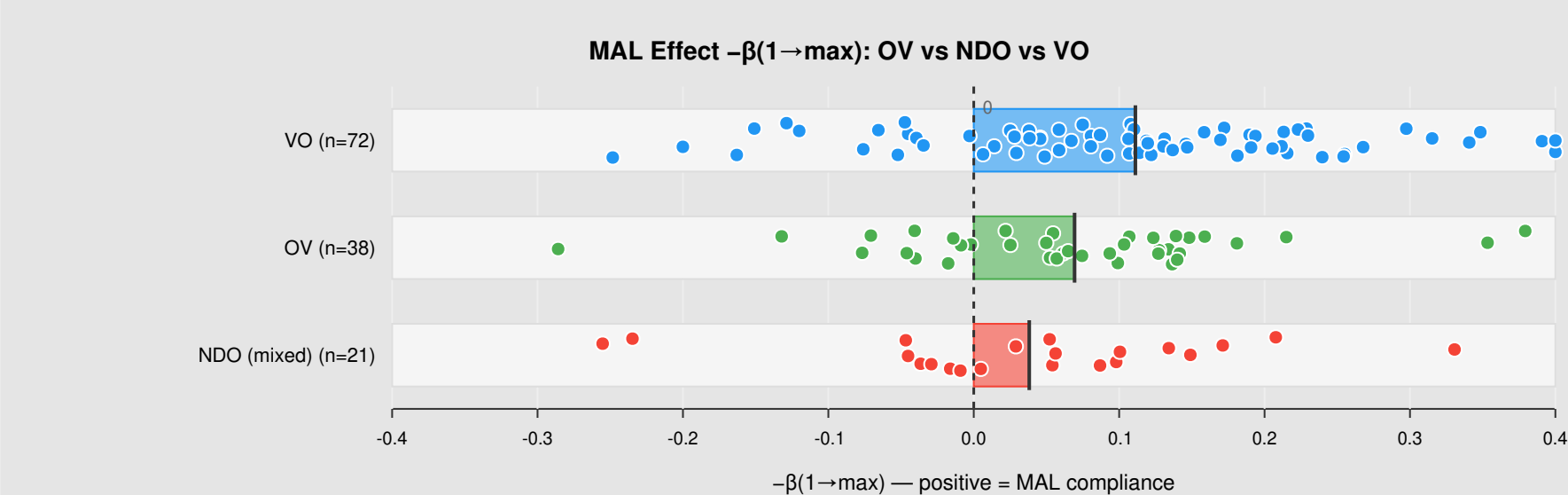
Naija – mixed, asymmetric
Left and right dependents diverge wildly. A young pidgincreole refusing to fit one curve – in fact the **strongest Anti-LMAL in the whole sample** ($\beta = +0.73$ preverbally).



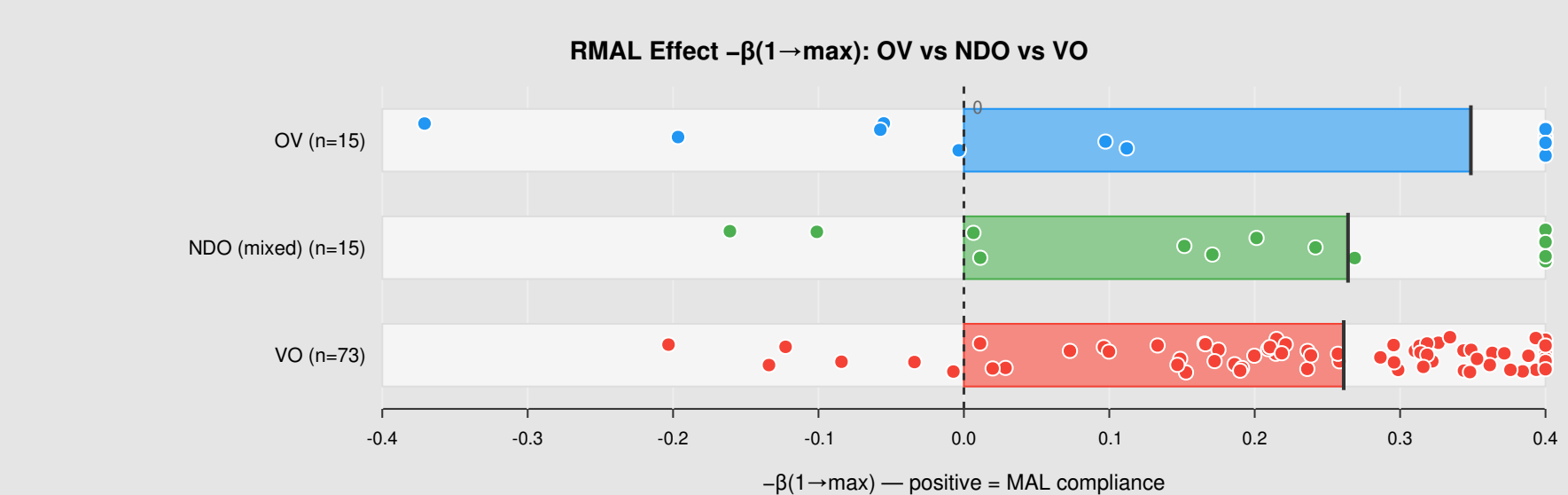
Arabic – complex behavior
Arabic has a particularly low R². It follows MAL from $n = 1$ to 3, but becomes Anti-MAL from $n = 3$ onwards.
And you? Check your language at typometrics.elizia.net!

THE NUMBERS NOBODY TALKS ABOUT

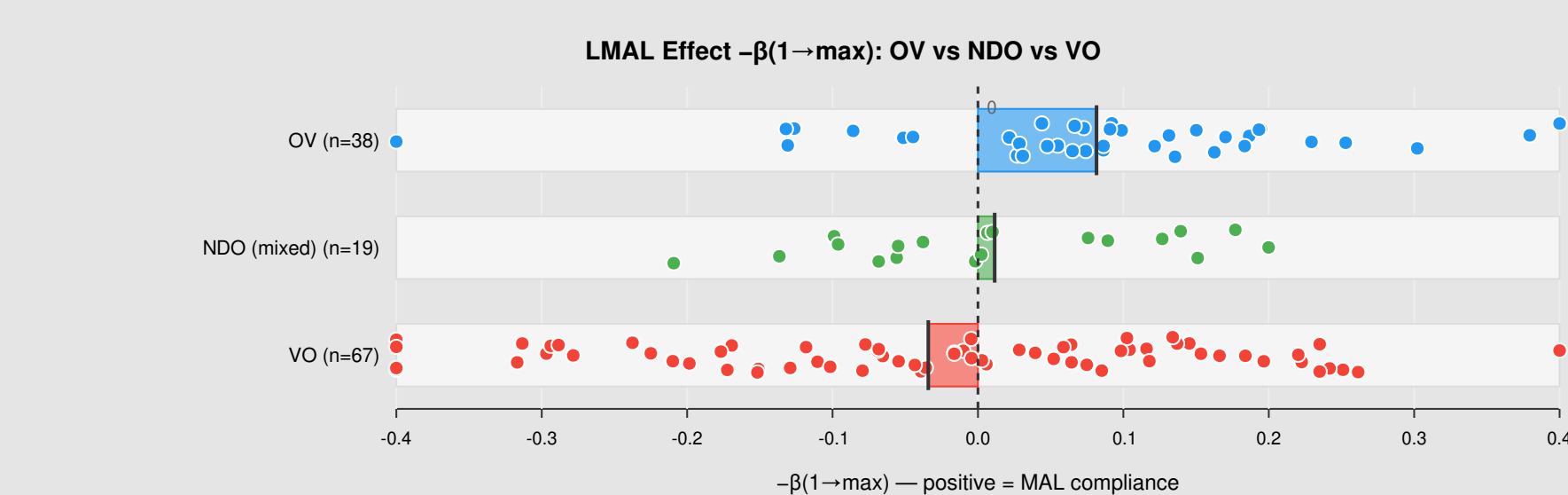
MAL is not ambidextrous. The verb leans right.



MAL by word order
MAL is stronger for **VO** languages (56%) than OV (42%), but RMAL and LMAL are **stronger for OV**! Where does the paradox come from?



RMAL: postverbal domain
79% of UD languages are RMAL (81/103) and only **7% are Anti-RMAL** (7/103). The right side of the verb strongly compresses constituent size as n grows. VO languages lead: 84% are RMAL vs. 63% for OV.



LMAL: preverbal domain
Only **31% are LMAL** (39/124) and **23% are Anti-LMAL** (29/124). The left side of the verb resists compression – or even reverses it. OV languages are better: 39% are LMAL vs. 29% for VO. VO languages are **3×** more likely to be Anti-LMAL.

IT'S 100% FREE*
Come and find out how your UD language is doing!